

GREAT BEND, KS, USA

WMO: 724517

Lat: 38.350N

Lon: 98.867W

Elev: 1880

StdP: 13.72

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	9.0	-5.9	4.3	6.6	-0.2	5.8	11.6	32.2	37.3	28.5	36.2	9.3	0	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	100.1	72.1	97.4	72.1	93.4	71.5	76.2	90.8	75.0	89.7	73.7	88.7	16.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	128.3	83.5	70.5	120.5	81.7	69.9	117.8	81.0	40.9	90.7	39.7	89.7	38.5	89.5	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.3	26.9	24.6	-2.7	104.8	4.5	2.9	-5.9	106.9	-8.5	108.5	-11.0	110.1	-14.3	112.2		
			-4.2	78.4	4.6	1.3	-7.5	79.3	-10.2	80.1	-12.8	80.8	-16.1	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	31.2	34.5	44.0	53.5	64.5	75.3	79.9	77.9	69.8	56.6	43.4	32.9
	DBStd	19.63	10.68	11.83	11.25	9.81	8.92	6.97	6.03	5.77	8.50	9.48	9.69	10.00
	HDD50	2173	584	446	246	73	5	0	0	0	1	43	241	534
	HDD65	5032	1048	855	652	361	123	8	1	48	289	650	996	
	CDD50	4142	1	12	61	178	453	758	926	865	596	247	42	3
	CDD65	1522	0	0	2	16	106	316	462	400	192	28	0	0
	CDH74	17417	0	8	52	282	1183	3596	5511	4336	2063	366	20	0
	CDH80	8671	0	2	5	84	467	1791	3040	2237	931	112	2	0
Wind	WSAvg	11.7	10.8	11.7	13.0	14.0	12.3	12.3	10.9	10.1	11.5	11.7	11.4	10.7
Precipitation	PrecAvg	26.5	0.7	0.9	1.6	2.3	4.4	3.7	3.5	3.7	1.9	1.8	1.1	0.8
	PrecMax	38.0	2.0	2.8	5.4	3.7	13.7	8.0	11.4	9.1	5.6	5.9	3.1	3.5
	PrecMin	16.7	0.0	0.0	0.0	0.3	1.0	1.1	0.0	0.3	0.3	0.1	0.1	0.0
	PrecStd	5.4	0.5	0.8	1.3	0.9	2.9	1.5	2.4	1.8	1.3	1.6	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.2	72.9	80.6	88.3	96.6	102.3	105.9	102.2	99.1	90.0	76.9	64.5
		MCWB	47.9	51.4	57.5	61.1	67.3	71.5	72.3	71.4	69.7	66.3	56.0	50.7
	2%	DB	59.1	65.7	73.4	81.6	90.2	97.3	101.5	99.1	93.2	82.3	70.5	58.8
		MCWB	44.2	49.5	56.1	61.0	67.5	72.2	72.4	72.2	69.8	62.3	53.7	46.1
	5%	DB	53.7	58.8	68.4	75.5	84.4	93.1	98.8	95.3	90.0	78.5	64.3	54.1
		MCWB	42.4	46.3	52.8	58.4	66.6	71.3	72.9	72.2	69.4	61.2	51.4	44.0
	10%	DB	46.5	53.7	63.3	71.7	81.1	90.3	93.5	91.2	85.2	72.7	60.6	48.1
		MCWB	38.6	43.5	51.6	57.4	65.3	70.5	72.7	71.6	67.5	59.4	50.1	40.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	55.4	63.6	66.7	72.6	77.0	78.2	77.8	74.5	70.2	59.4	54.3
		MCDB	62.3	64.8	71.3	78.8	85.2	92.2	93.6	91.1	88.2	80.7	69.2	60.0
	2%	WB	45.5	50.9	59.4	64.0	70.7	75.1	76.4	75.8	72.6	66.6	56.7	48.4
		MCDB	57.0	62.5	65.8	74.5	83.8	90.1	91.4	89.8	87.1	76.9	65.3	56.9
	5%	WB	42.4	47.1	55.9	61.2	68.8	73.5	75.2	74.4	70.9	63.8	53.4	44.3
		MCDB	52.2	58.9	66.4	72.8	81.1	88.2	90.1	88.9	85.1	72.4	62.6	53.6
	10%	WB	39.3	43.5	51.9	58.3	67.0	72.0	74.1	73.0	69.2	60.8	50.3	40.7
		MCDB	47.0	53.5	63.3	70.3	77.9	86.3	89.5	87.6	82.0	71.0	60.3	48.2
Mean Daily Temperature Range	5% DB	MDBR	22.5	23.3	24.8	24.7	22.9	23.6	23.8	23.3	24.0	24.8	24.6	21.9
		MCDBR	32.9	34.4	34.1	32.3	29.3	27.9	29.5	28.7	28.2	31.5	32.5	30.8
		MCWBR	21.0	20.8	19.3	17.4	12.6	10.1	9.2	9.7	10.7	15.8	19.0	20.1
	5% WB	MCDBR	30.7	32.1	27.7	27.6	24.2	24.7	24.1	23.9	24.0	24.5	28.3	28.6
		MCWBR	20.3	20.2	17.3	16.8	12.3	10.7	9.7	10.0	10.4	14.2	17.6	19.6
Clear-Sky Solar Irradiance	taub	0.274	0.284	0.318	0.342	0.369	0.391	0.411	0.397	0.363	0.322	0.282	0.271	
	taud	2.527	2.502	2.425	2.391	2.398	2.363	2.337	2.366	2.408	2.500	2.542	2.528	
	Ebn at Noon	292	303	300	297	289	281	275	276	280	284	286	285	
	Edh at Noon	24	28	34	37	37	39	40	37	34	28	23	22	
All-Sky Solar Radiation	RadAvg	801	1053	1407	1747	2017	2248	2194	1953	1617	1184	884	702	
	RadStd	70	118	110	98	136	126	105	113	98	128	82	58	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.17	+1.03	N/A	N/A	N/A	N/A	
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page